

Used Car Market Analysis Dashboard using SQL and Power BI

Project Objective

The main objective of this project is:

- Analyze used car market trends
- Find most popular car brands and fuel types
- Identify factors affecting selling price
- Create interactive Power BI dashboards
- Perform complete data analysis using SQL queries



Dataset Information

The CarDekho dataset contains information about used cars listed for sale, including car brand, model, fuel type, transmission type, vehicle age, mileage, engine capacity, kilometers driven, and selling price. The dataset is used for analyzing car market trends, resale value, and customer preferences.

Tools used

- SQL
- POWER BI

PROJECT WORKFLOW IN SQL

- Step 1- Create database in sql
- Step 2- Import the csv file
- Step 3- Data cleaning in sql

1. **Check null values** : `SELECT * FROM cardekho_dataset WHERE car_name IS NULL OR brand IS NULL OR selling_price IS NULL;`

2. **Remove Duplicate Rows** : `SELECT car_name, COUNT(*) FROM cardekho_dataset GROUP BY car_name HAVING COUNT(*) > 1;`

3. **Check Unique Brands** : `SELECT DISTINCT brand FROM cardekho_dataset;`

Result link =

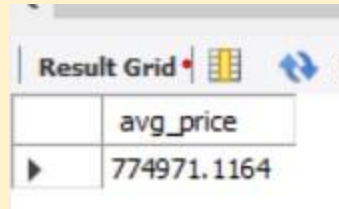
https://docs.google.com/spreadsheets/d/e/2PACX-1vTcEfMY6Kgd51lyC7Nbue_dX9BNvqA9wSRIh-vU9uHJhUXFhFQ3AjjUlly0UM9tCSyl9JY55euSdmUA/pub?gid=1295794155&single=true&output=pdf

Step 4:

Exploratory Data Analysis (EDA) in SQL

1. Average Selling Price : **SELECT AVG(selling_price) AS avg_price FROM cardekho_dataset;**

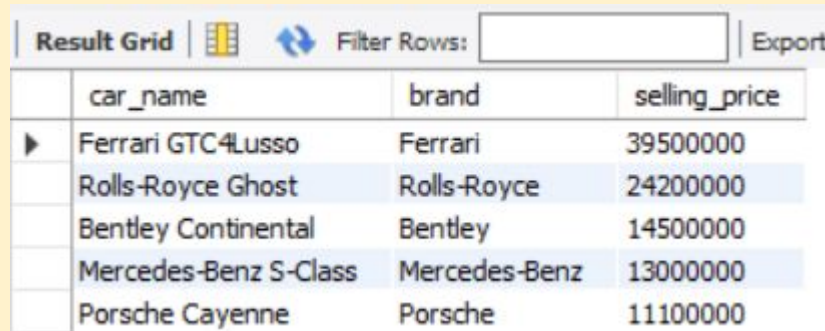
Result ;



avg_price
774971.1164

2. Most Expensive Cars : **SELECT car_name, brand, selling_price FROM cardekho_dataset ORDER BY selling_price DESC LIMIT 5;**

Result ;



car_name	brand	selling_price
Ferrari GTC4Lusso	Ferrari	39500000
Rolls-Royce Ghost	Rolls-Royce	24200000
Bentley Continental	Bentley	14500000
Mercedes-Benz S-Class	Mercedes-Benz	13000000
Porsche Cayenne	Porsche	11100000

3. Cheapest Cars ;`SELECT car_name, brand, selling_price FROM cardekho_dataset ORDER BY selling_price ASC LIMIT 5;`

Result;

Result Grid   Filter Rows:			
	car_name	brand	selling_price
▶	Maruti Wagon R	Maruti	40000
	Maruti Alto	Maruti	45000
	Maruti Wagon R	Maruti	50000
	Maruti Alto	Maruti	50000
	Honda City	Honda	50000

4. Top Brands by Number of Cars ;`SELECT brand, COUNT(*) AS total_cars FROM cardekho_dataset GROUP BY brand ORDER BY total_cars DESC Limit 3;`

Result;

Result Grid   Filter		
	brand	total_cars
▶	Maruti	4992
	Hyundai	2982
	Honda	1485

Step 5 ;

Intermediate SQL Queries

1.Cars Above Average Selling Price ; `SELECT car_name, brand, selling_price FROM cardekho_dataset WHERE selling_price > ( SELECT AVG(selling_price) FROM cardekho_dataset );`

Results; <https://docs.google.com/spreadsheets/d/e/2PACX-1vRB94clSgIb-b-sphf3gyy-YfZVpHEf7j-mkmAOtl0ysThkiwGUX3FJYFDyDJtXtkcPX9jgeElbmE7W/pub?output=pdf>

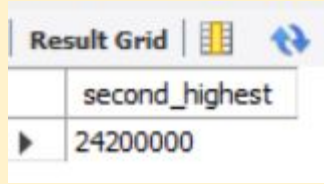
2.Brand with Highest Average Price ; `SELECT brand, ROUND(AVG(selling_price),2) AS avg_price FROM cardekho_dataset GROUP BY brand ORDER BY avg_price DESC LIMIT 1;`

Results;

Result Grid			Filter Row
	brand	avg_price	
▶	Ferrari	39500000.00	

3.Find Second Highest Selling Price ;`SELECT MAX(selling_price) AS second_highest FROM cardekho_dataset WHERE selling_price < ( SELECT MAX(selling_price) FROM cardekho_dataset );`

Results;

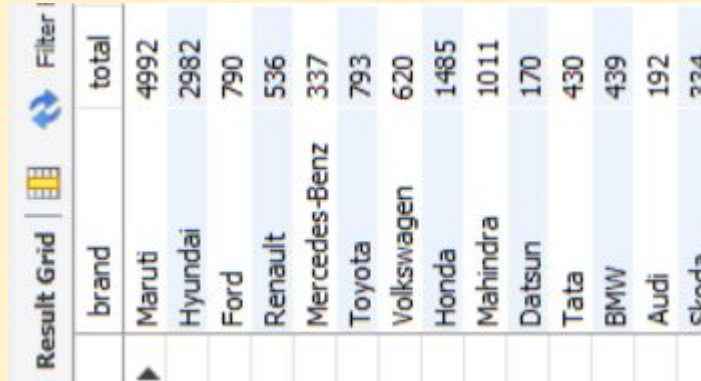


A screenshot of a database result grid. The grid has two columns: 'second_highest' and a value '24200000'. The grid is titled 'Result Grid' and has a refresh icon.

second_highest
24200000

4.Cars Having Same Brand Multiple Times ;`SELECT brand, COUNT(*) AS total FROM cardekho_dataset GROUP BY brand HAVING COUNT(*) > 100;`

Results;



A screenshot of a database result grid showing car brands and their total counts. The grid has two columns: 'brand' and 'total'. The grid is titled 'Result Grid' and has a refresh icon.

brand	total
Maruti	4992
Hyundai	2982
Ford	790
Renault	536
Mercedes-Benz	337
Toyota	793
Volkswagen	620
Honda	1485
Mahindra	1011
Datsun	170
Tata	430
BMW	439
Audi	192
Skoda	334

